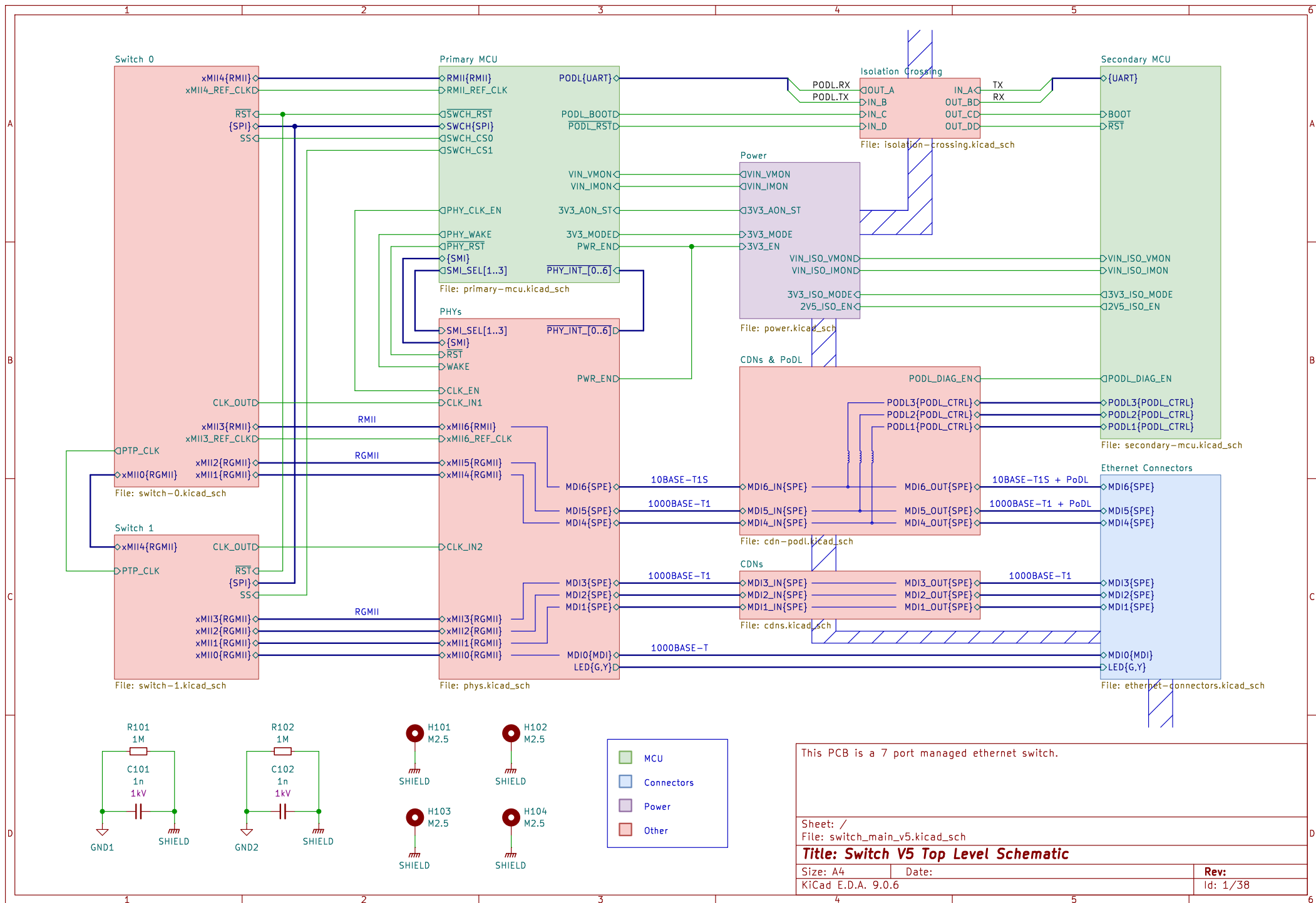
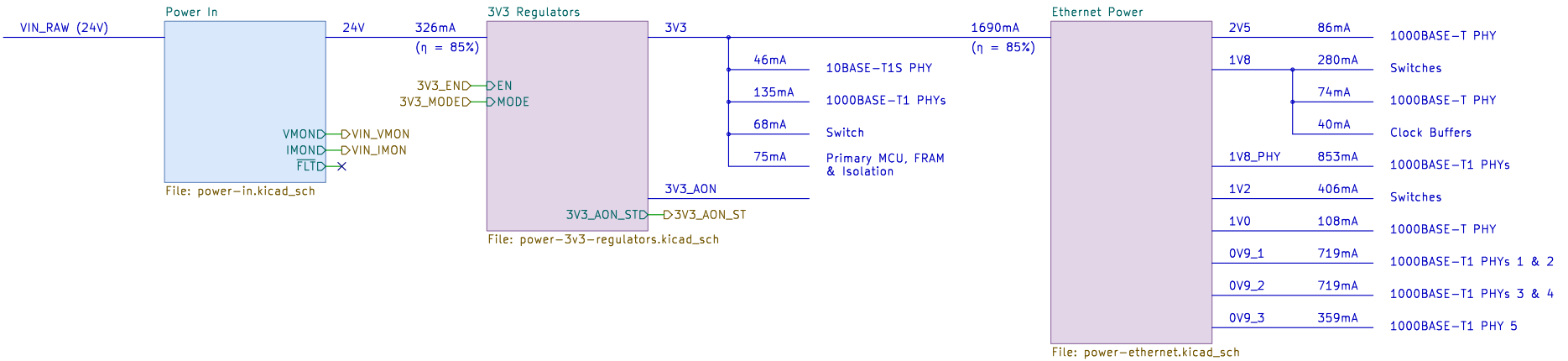


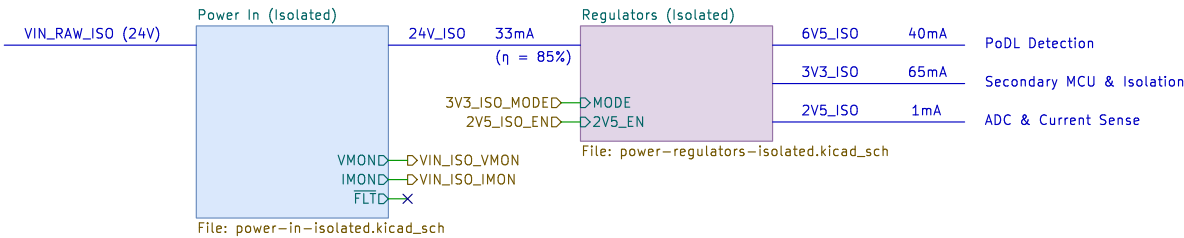


PHY Side Power



Isolation Barrier

Connector Side Power



Block diagram of the power supply scheme for both sides of the isolation barrier.

Sheet: /Power/
File: power.kicad_sch

Title: Power

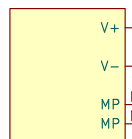
Size: A4
KiCad E.D.A. 9.0.6

Date:

Rev:

Id: 2/38

J301
MKXS1-1 Power Input



ESD
Protection

Surge
Protection

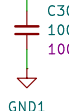
VIN_RAW

D301
SZESD7241N2T5G



U302
TVS3301DRBR

GND1



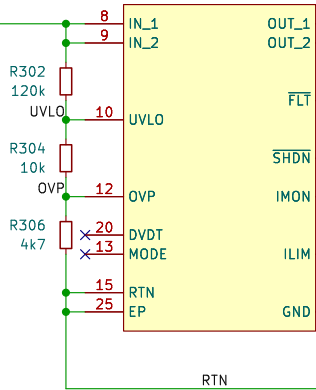
C301
100n
100V

GND1

UVLO = 10.9V
OVP = 34.1V

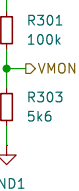
eFuse

U301
TPS26600RHFR



$V_{MON} = V_{IN} / 18.857$
The 150mΩ eFuse
resistance should be
taken into account when
calculating input power.

+VIN



$I_{OUT} = V_{MON} / 7.84$

$R_{ILIM} = 22k\Omega$
 $\therefore I_{LIM} = 12 / 22 = 545mA$

TPS26600 has auto-retry
when a fault occurs.

Power input and protection for the PHY side of the PCB.

Sheet: /Power/Power In/
File: power-in.kicad_sch

Title: Power In (PHY Side)

Size: A4

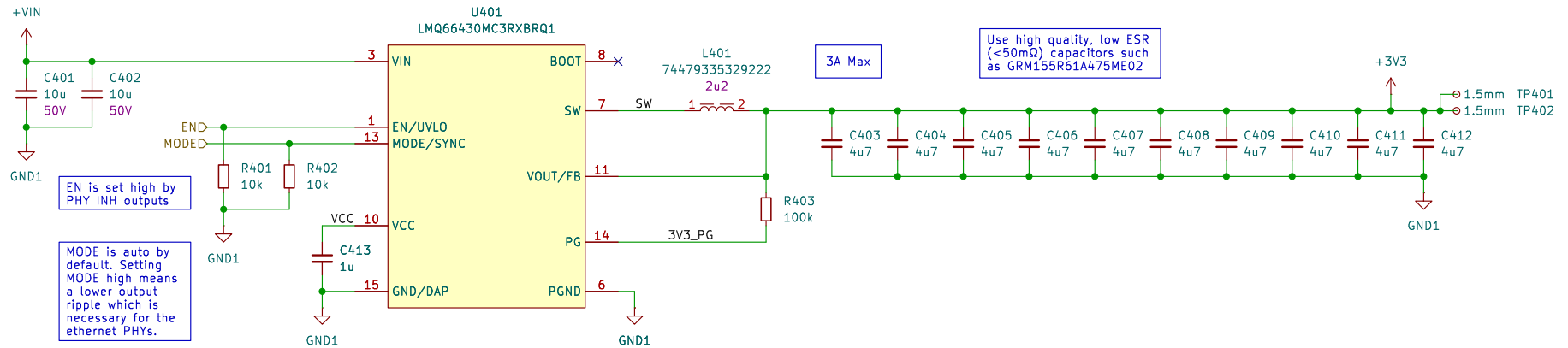
Date:

KiCad E.D.A. 9.0.6

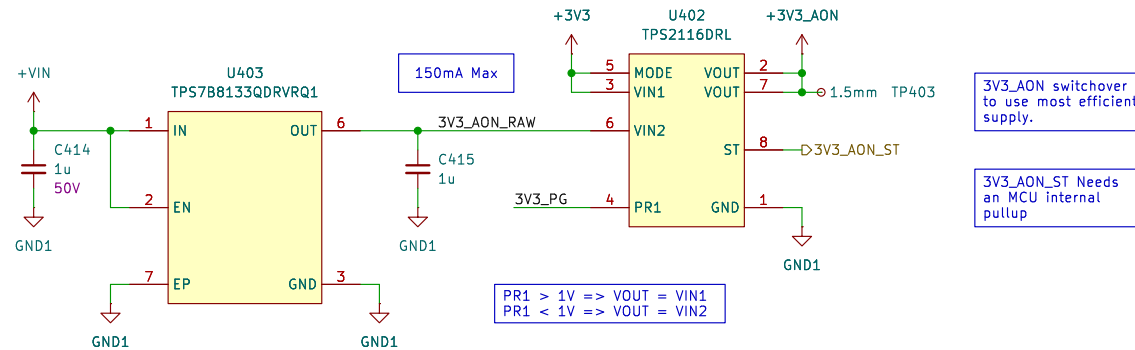
Rev:

Id: 3/38

3V3 Main Regulator



3V3 Always on supply



3V3 and 3V3_AON (always on) voltage regulators.

Sheet: /Power/3V3 Regulators/
File: power-3v3-regulators.kicad_sch

Title: 3V3 Regulators (PHY Side)

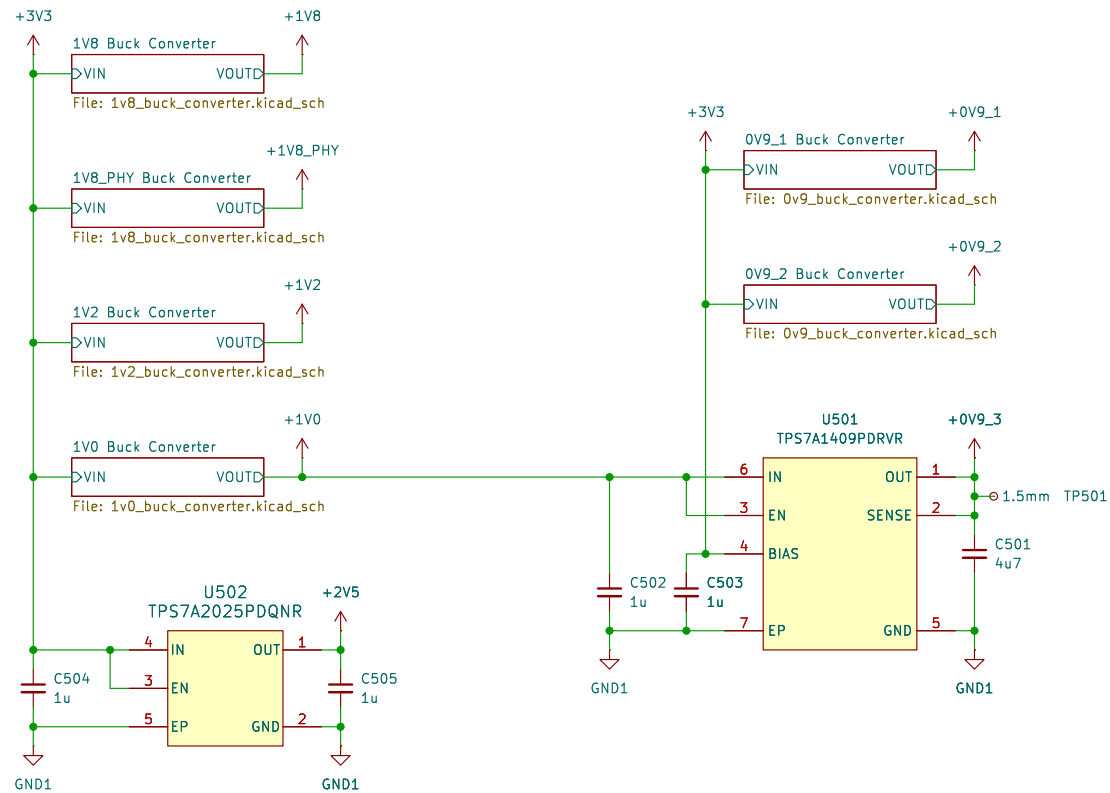
Size: A4

Date:

Rev:

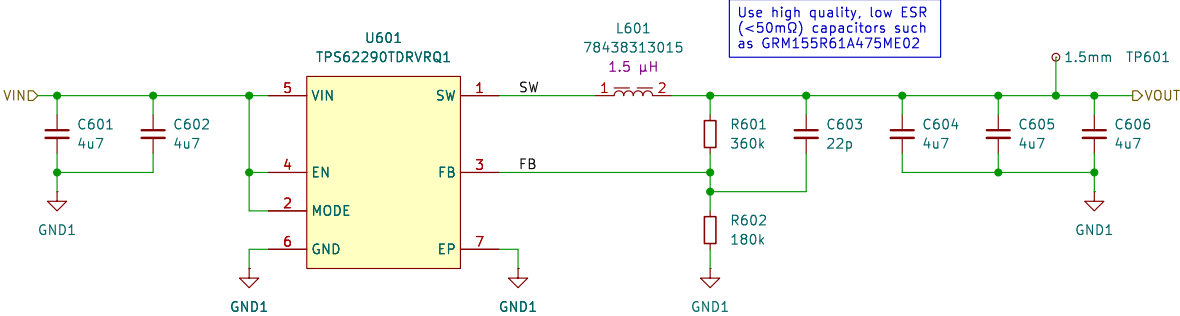
KiCad E.D.A. 9.0.6

Id: 4/38



Power supplies for ethernet switch and PHYs.		
Sheet: /Power/Ethernet Power/ File: power-ethernet.kicad_sch		
Title: Ethernet Power (PHY Side)		
Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.6		Id: 5/38

1V8 Regulator



1.8V, 1A buck SMPS.

Sheet: /Power/Ethernet Power/1V8 Buck Converter/
File: 1v8_buck_converter.kicad_sch

Title:

Size: A4

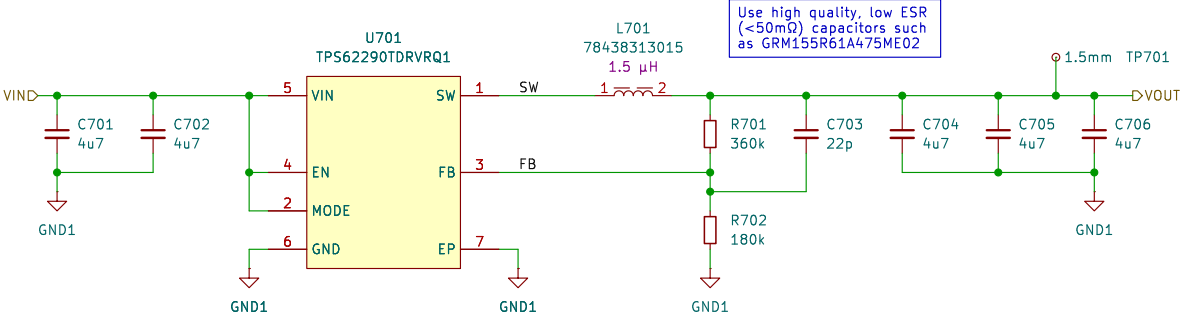
Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 6/38

1V8 Regulator



1.8V, 1A buck SMPS.

Sheet: /Power/Ethernet Power/1V8_PHY Buck Converter/
File: 1v8_buck_converter.kicad_sch

Title:

Size: A4

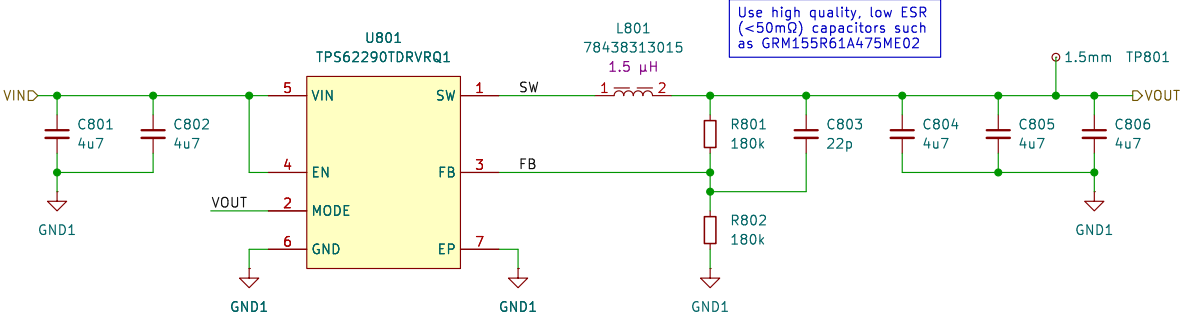
Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 7/38

1V2 Regulator



1.2V, 1A buck SMPS.

Sheet: /Power/Ethernet Power/1V2 Buck Converter/
File: 1v2_buck_converter.kicad_sch

Title:

Size: A4

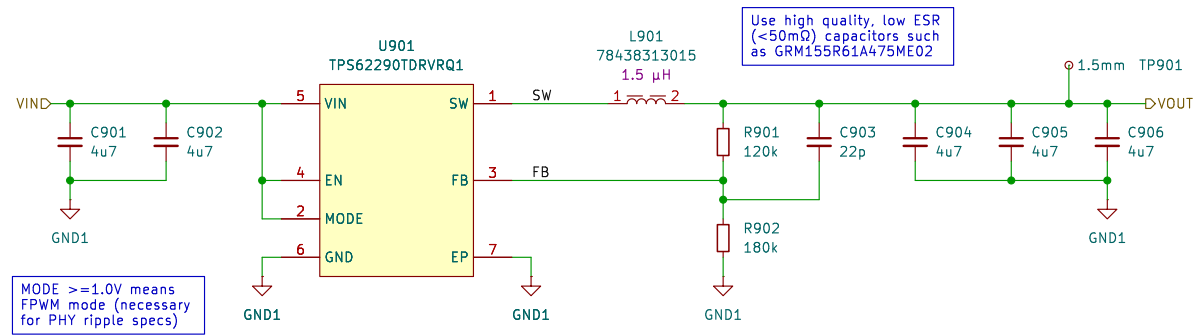
Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 8/38

1V0 Regulator



1.0V, 1A buck SMPS.

Sheet: /Power/Ethernet Power/1V0 Buck Converter/
File: 1v0_buck_converter.kicad_sch

Title:

Size: A4

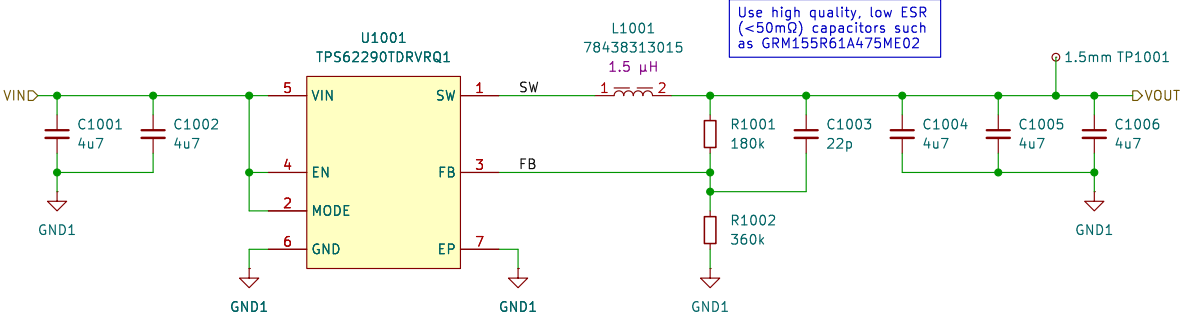
Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 9/38

0V9 Regulator



0.9V, 1A buck SMPS.

Sheet: /Power/Ethernet Power/0V9_1 Buck Converter/
File: 0v9_buck_converter.kicad_sch

Title:

Size: A4

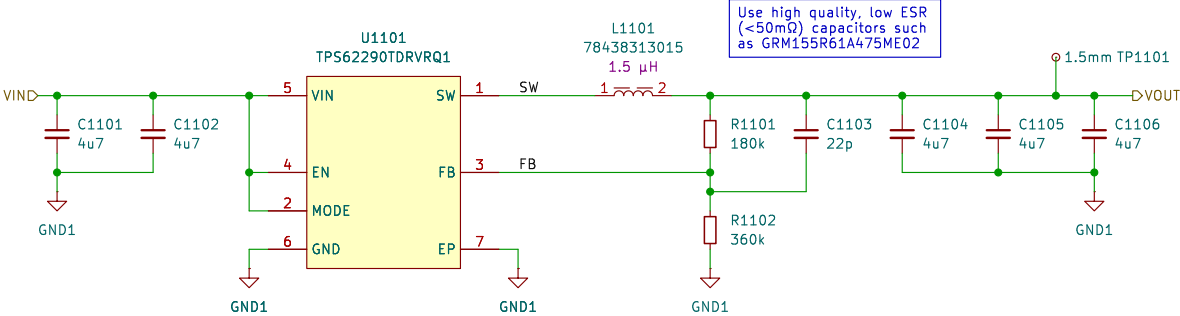
Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 10/38

0V9 Regulator



0.9V, 1A buck SMPS.

Sheet: /Power/Ethernet Power/0V9_2 Buck Converter/
File: 0v9_buck_converter.kicad_sch

Title:

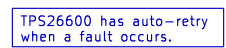
Size: A4

Date:

KiCad E.D.A. 9.0.6

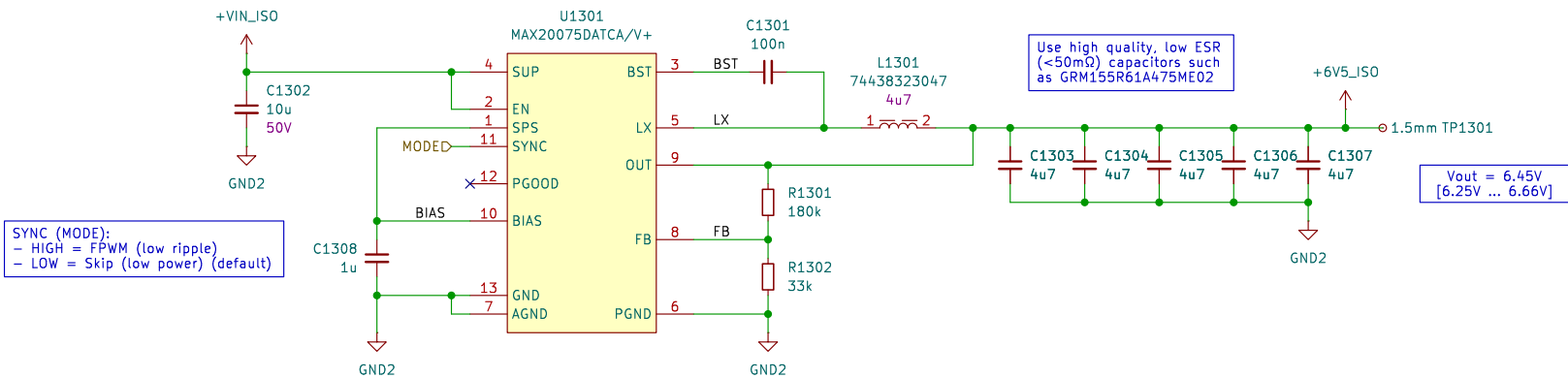
Rev:

Id: 11/38

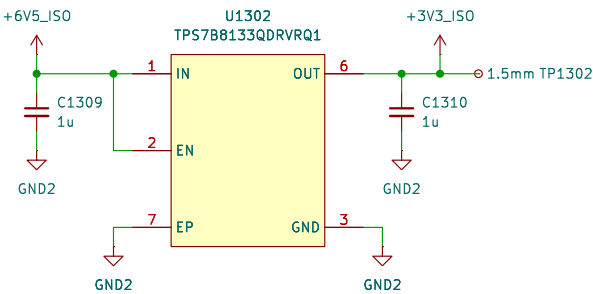


Id: 12/38

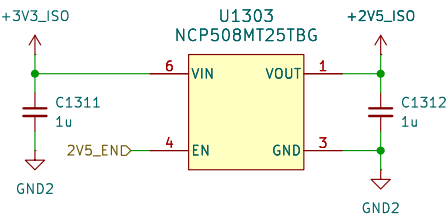
6V5 Regulator



3V3 Regulator



2V5 Regulator



6V5, 3V3 and 2V5 regulators for the connector side of the PCB.

Sheet: /Power/Regulators (Isolated)/
File: power-regulators-isolated.kicad_sch

Title: Regulators (Connector Side)

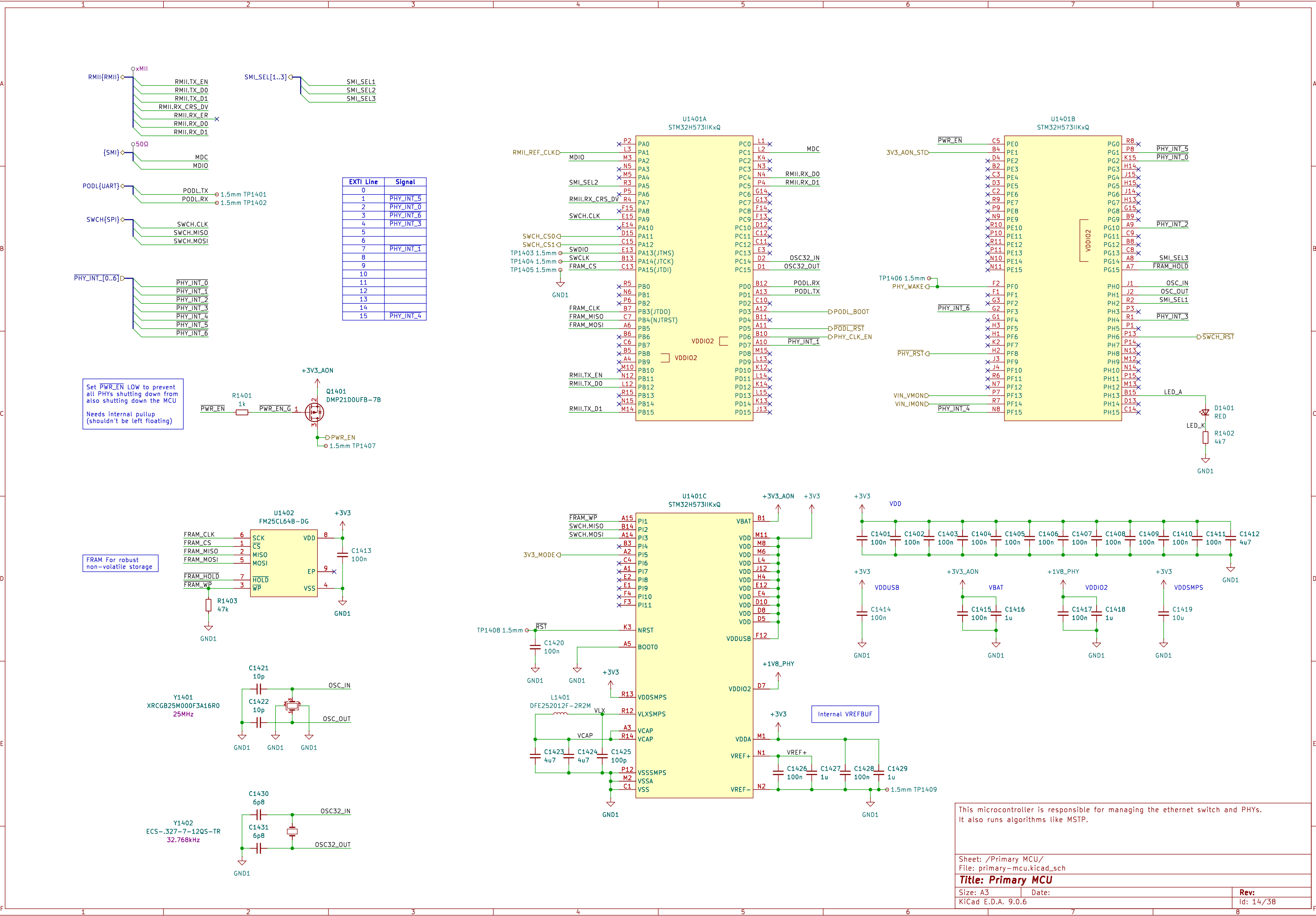
Size: A4

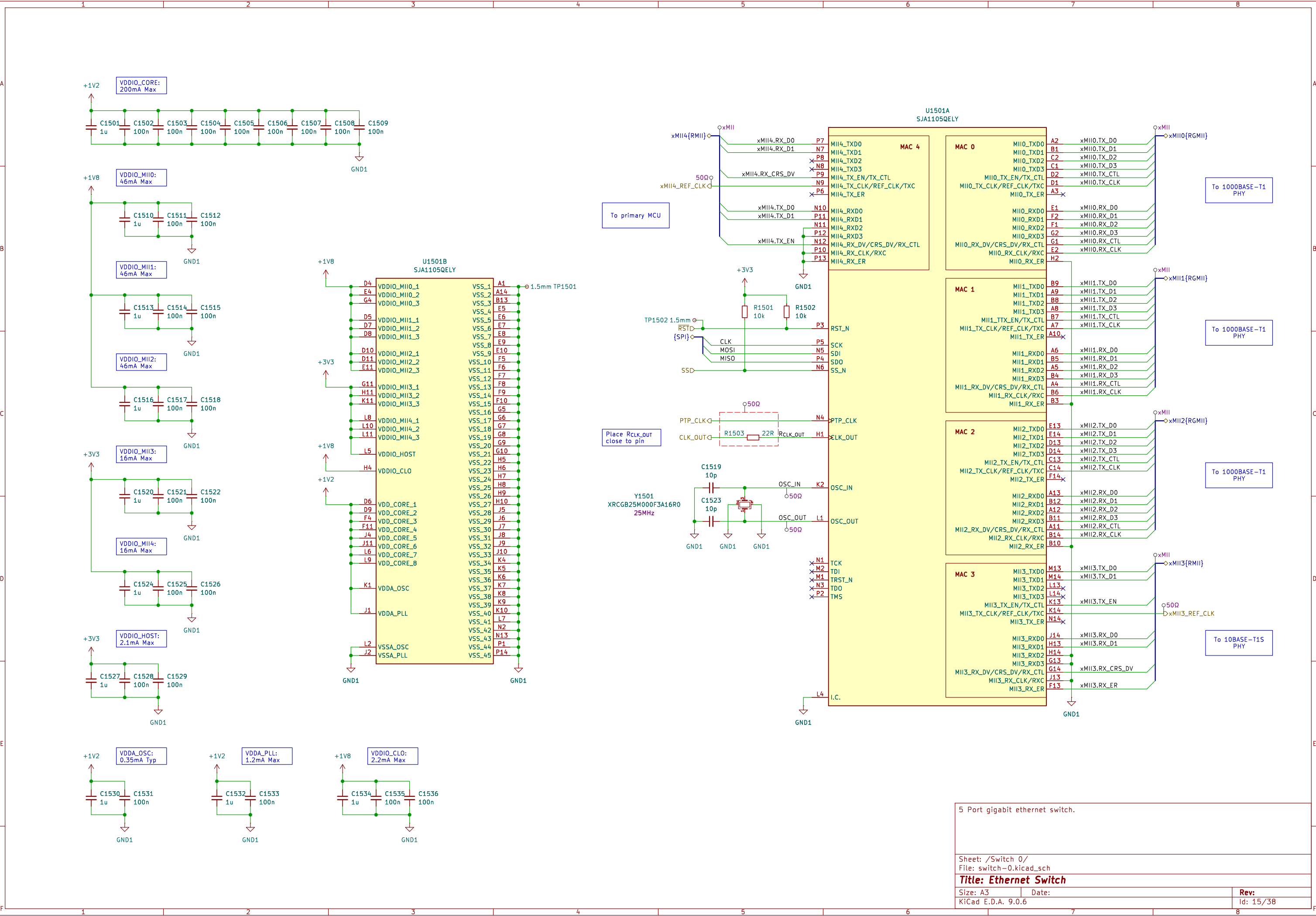
Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 13/38





5 Port gigabit ethernet switch.

Sheet: /Switch 0/
File: switch-0.kicad_sch

Title: Ethernet Switch

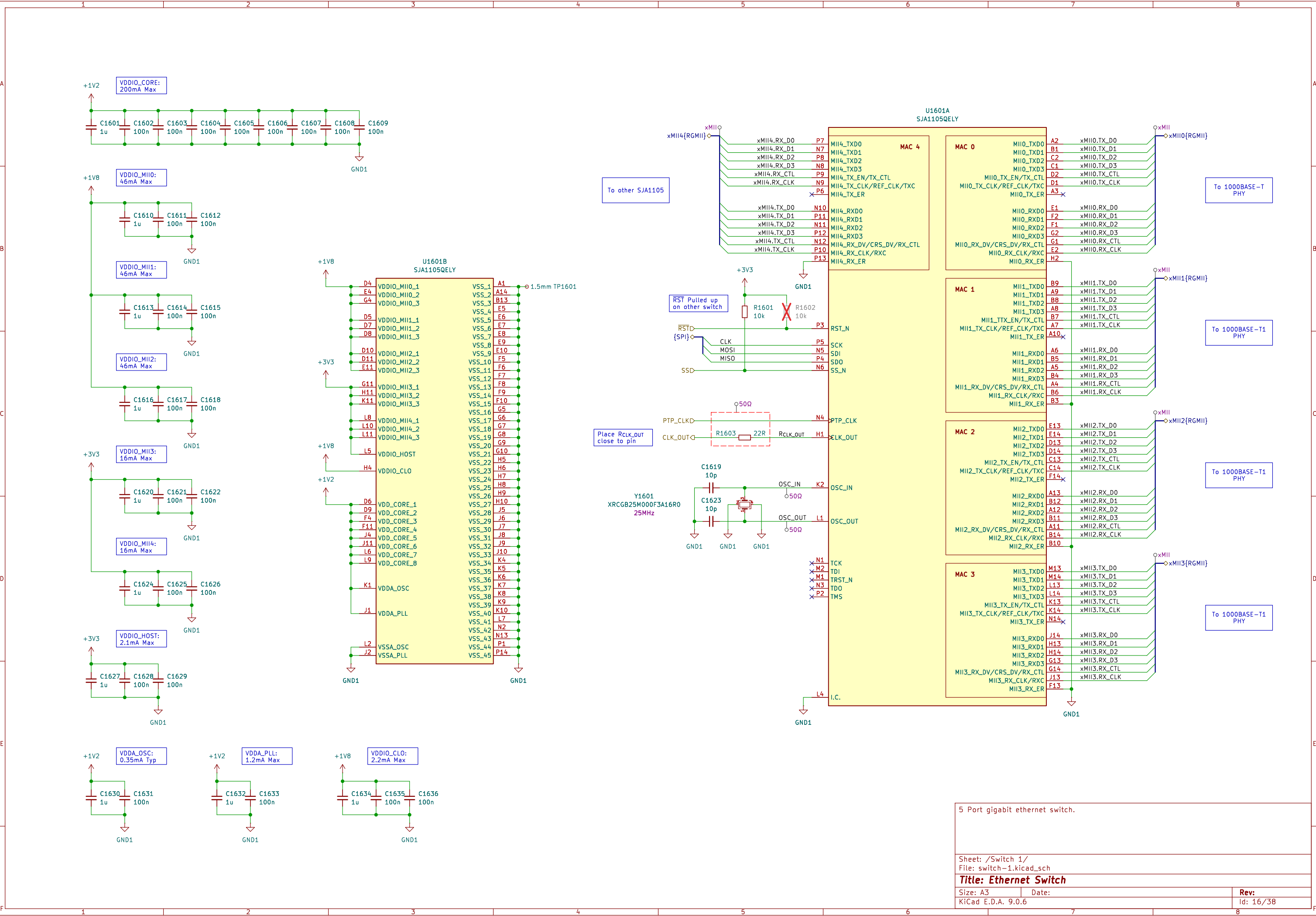
Size: A3

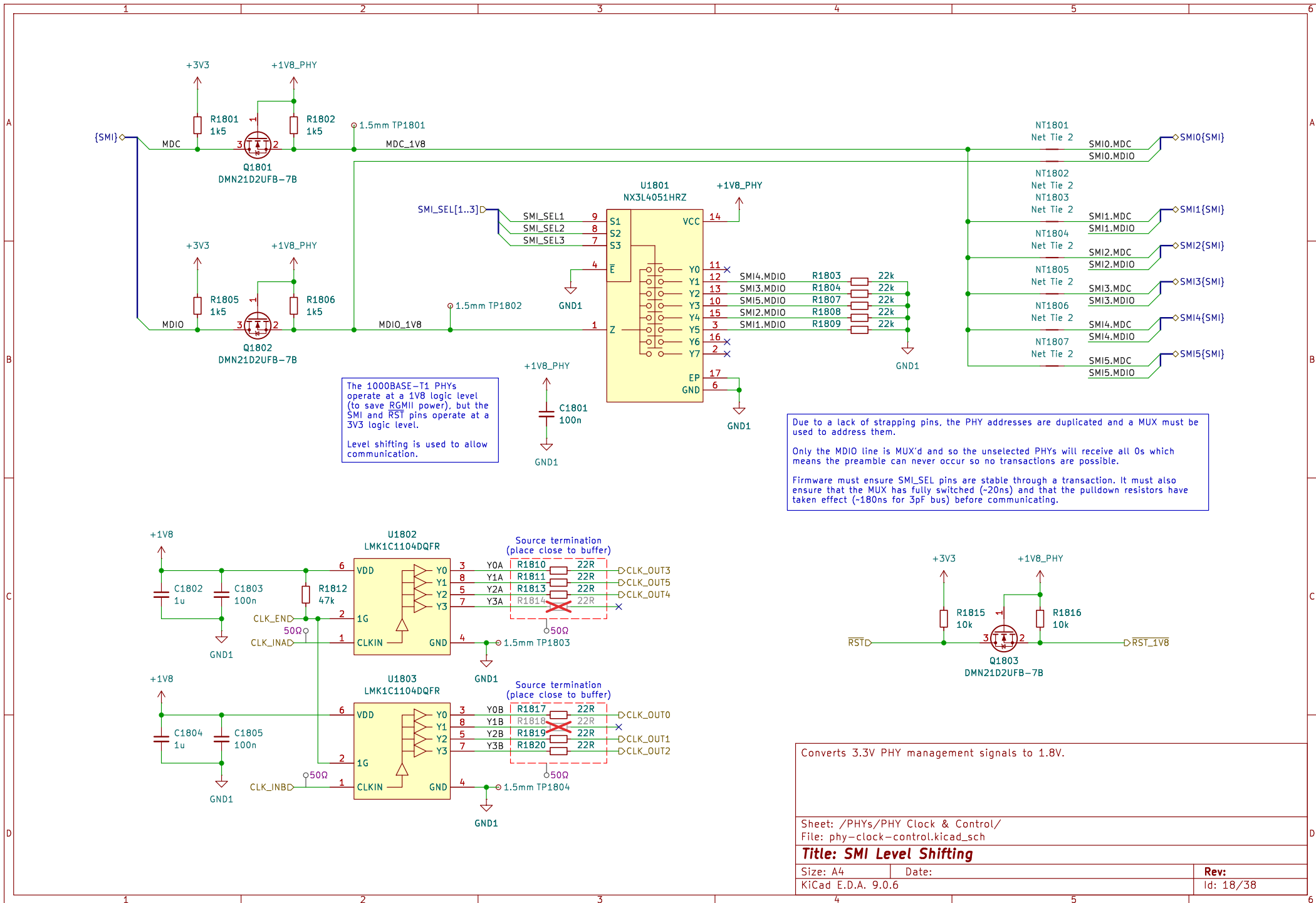
Date:

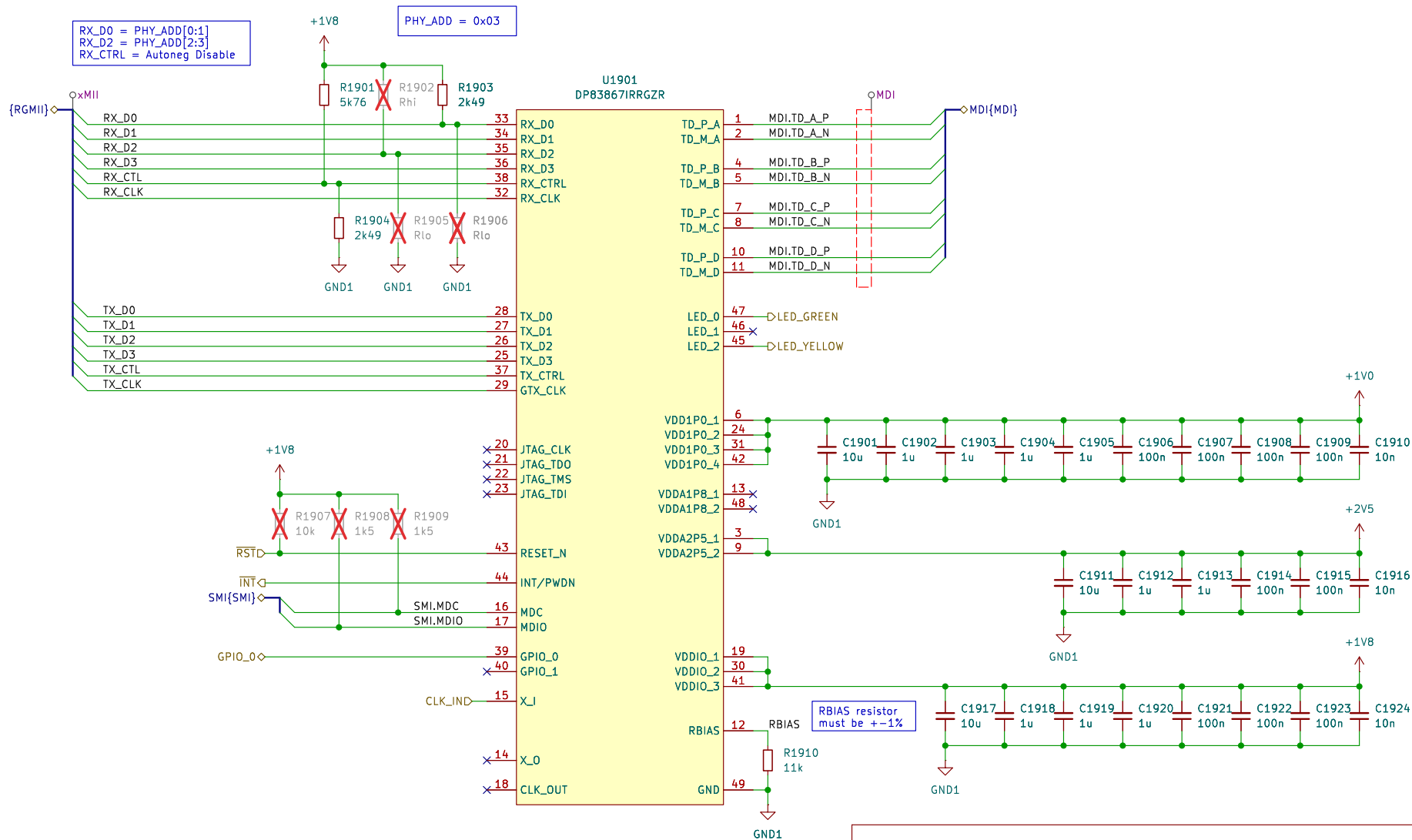
Rev:

KiCad E.D.A. 9.0.6

Id: 15/38







Sheet: /PHYs/1000BASE-T PHY/
File: 1000base-t-phy.kicad_sch

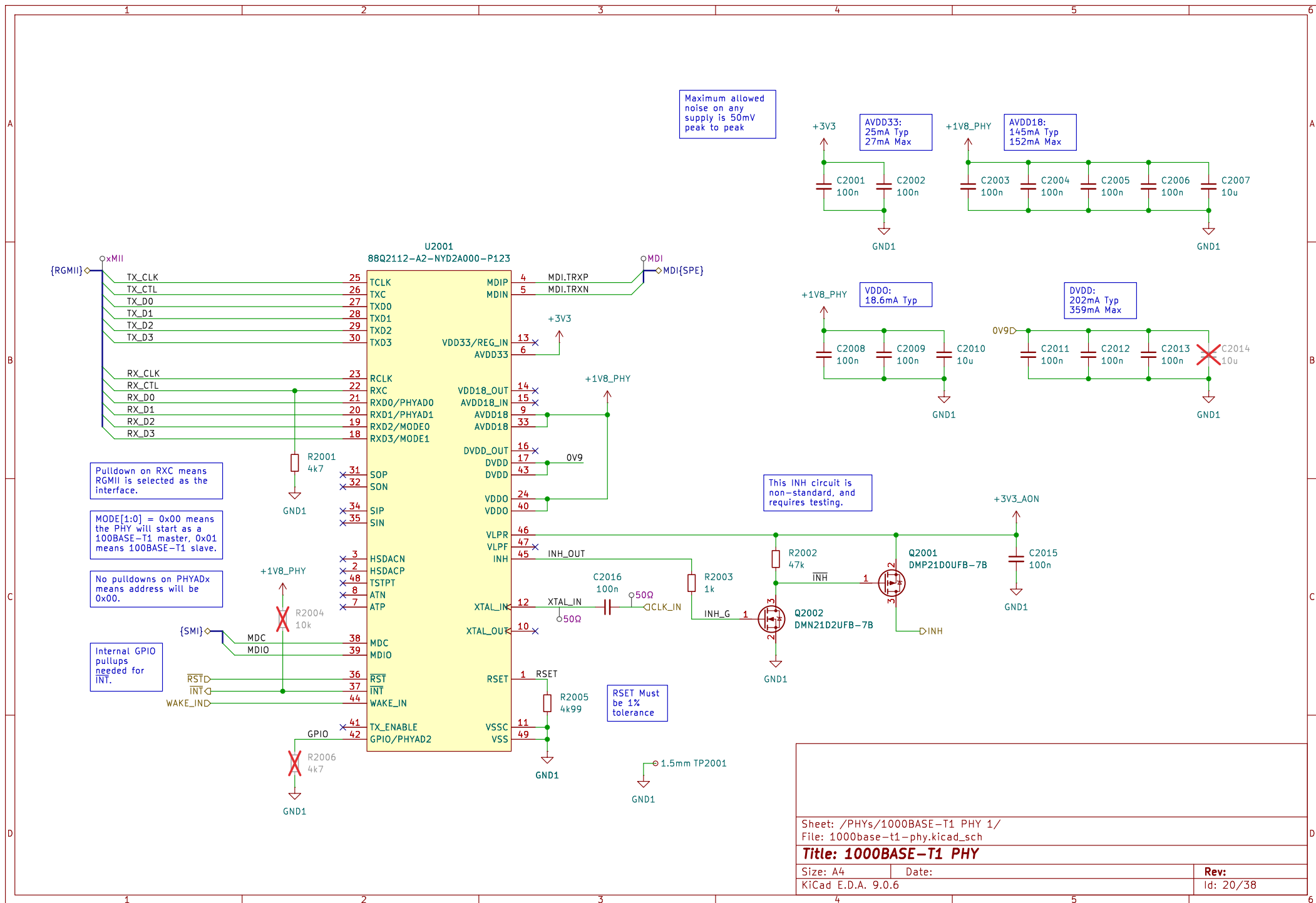
Title: PHY

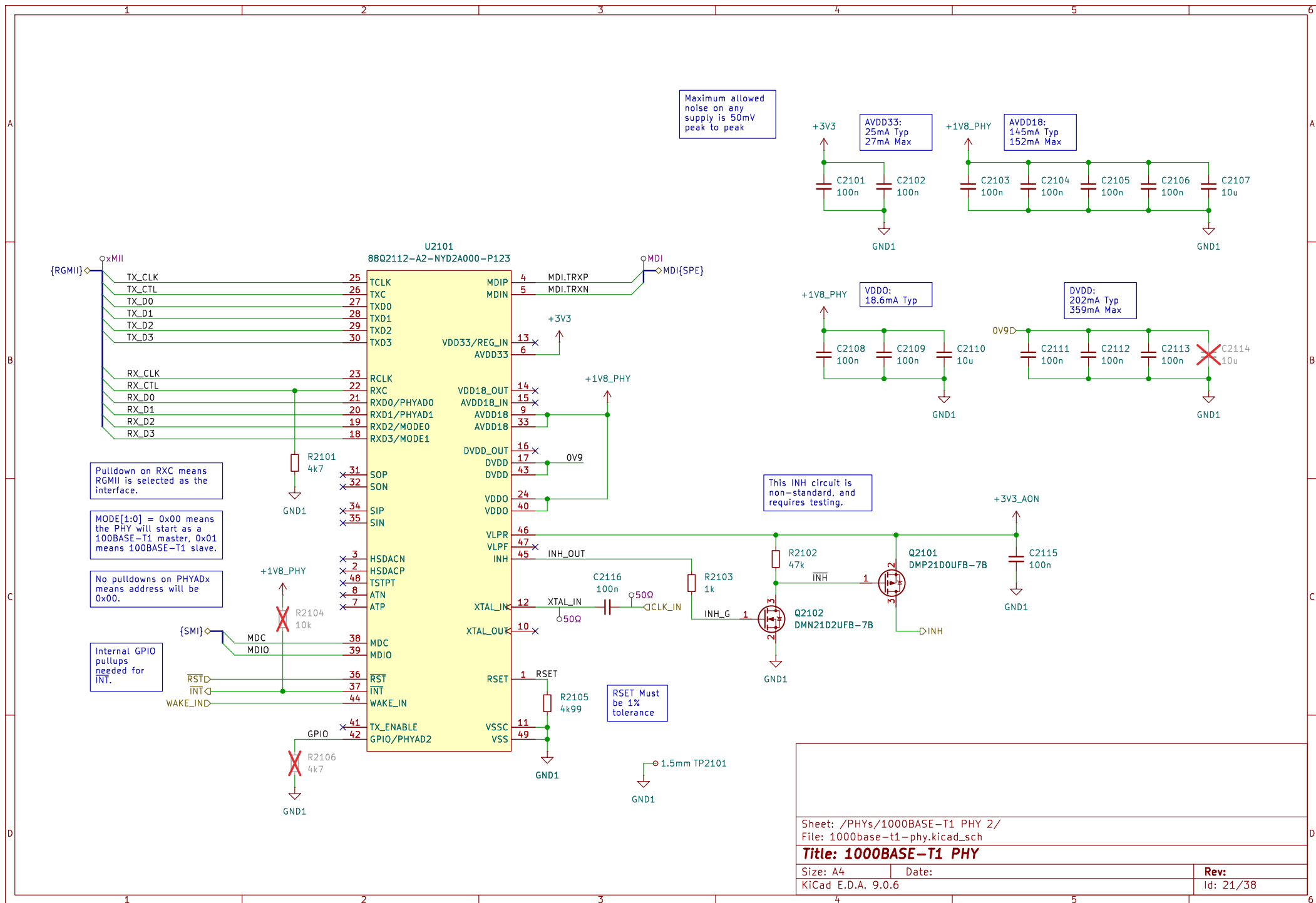
Size: A4 Date:

KiCad E.D.A. 9.0.6

Rev: 1.0.0

Id: 19/38





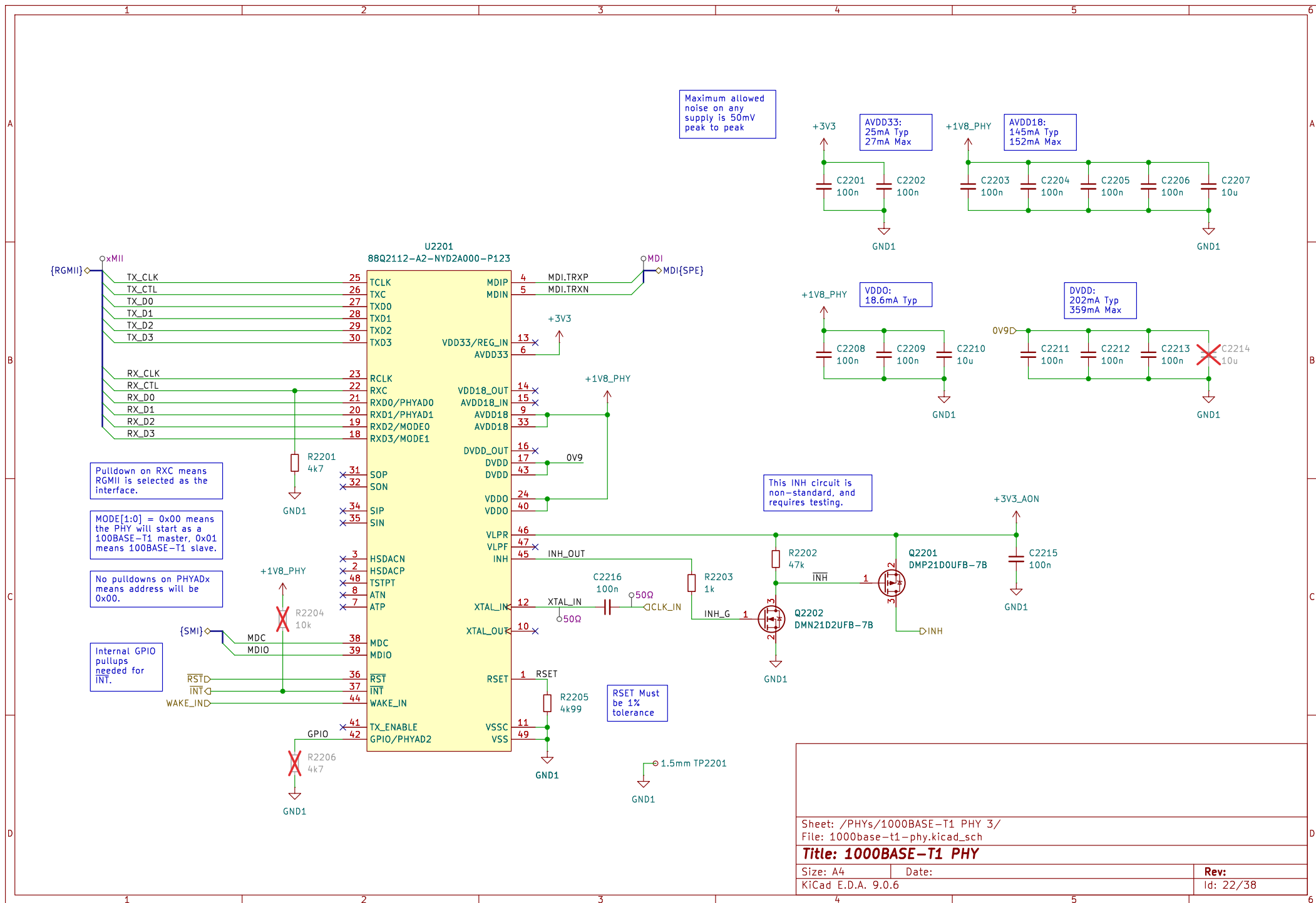
Sheet: /PHYs/1000BASE-T1 PHY 2/
File: 1000base-t1-phy.kicad_sch

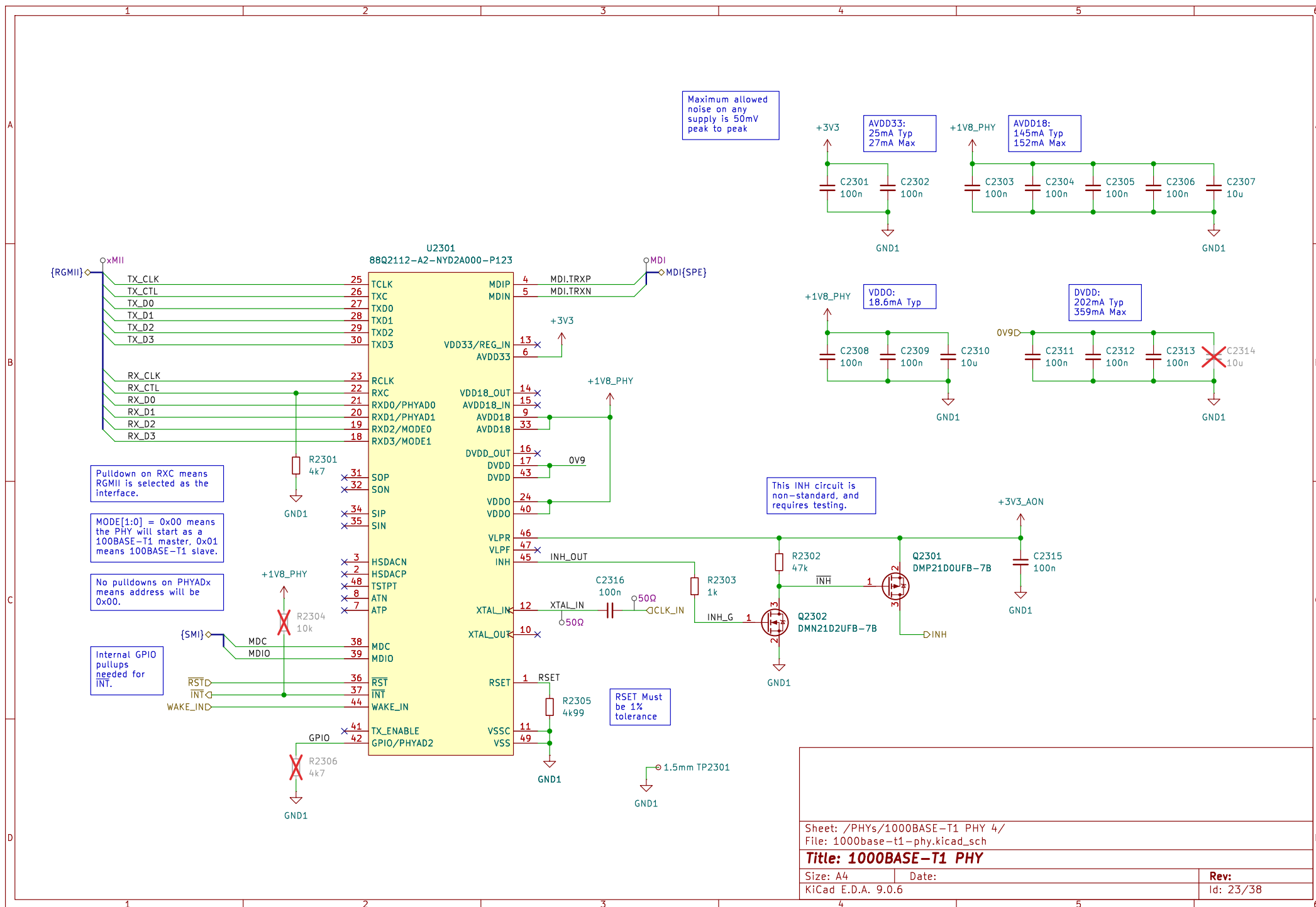
Title: 1000BASE-T1 PHY

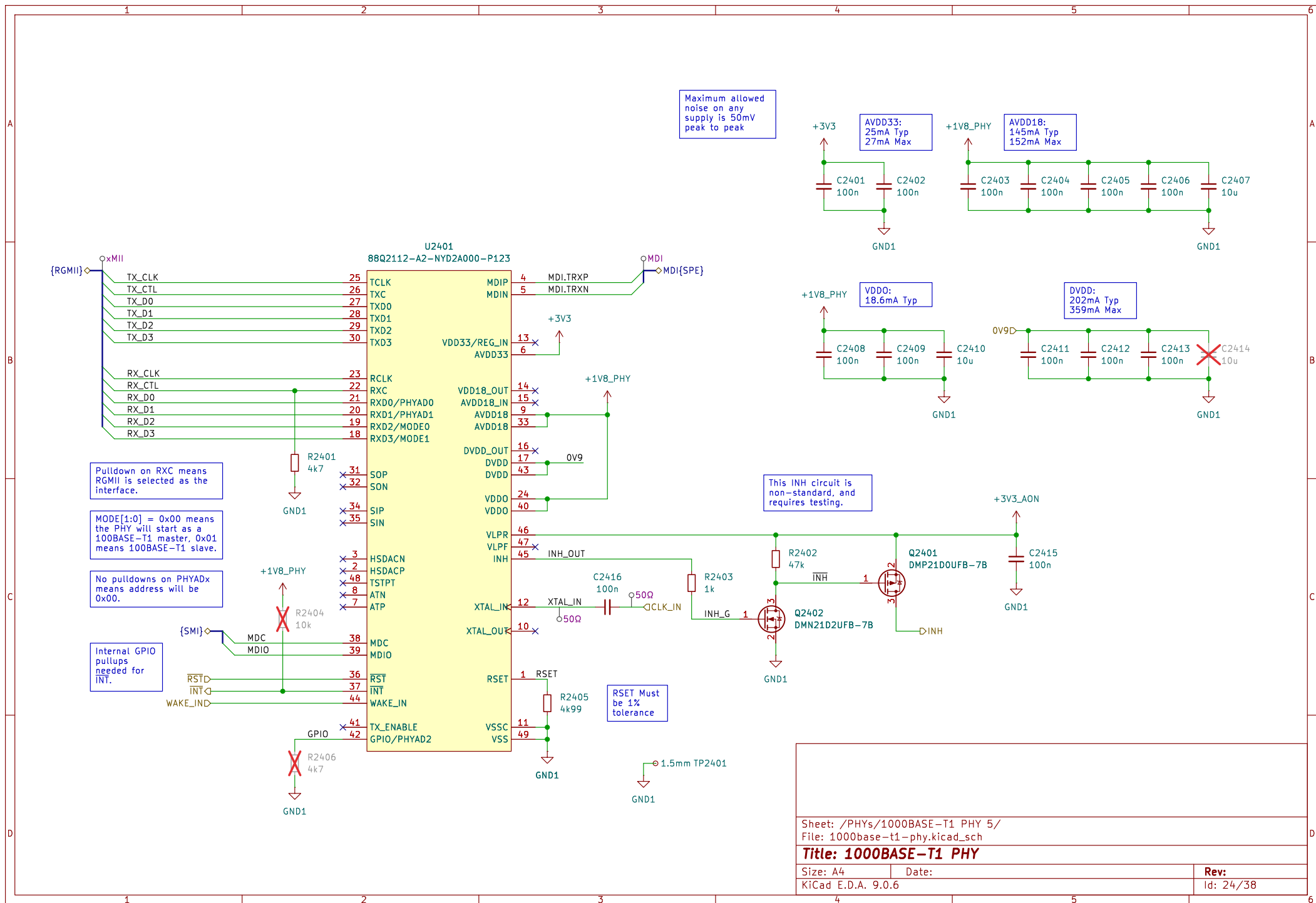
Size: A4
KiCad E.D.A. 9.0.6

Date:

Rev:
Id: 21/38







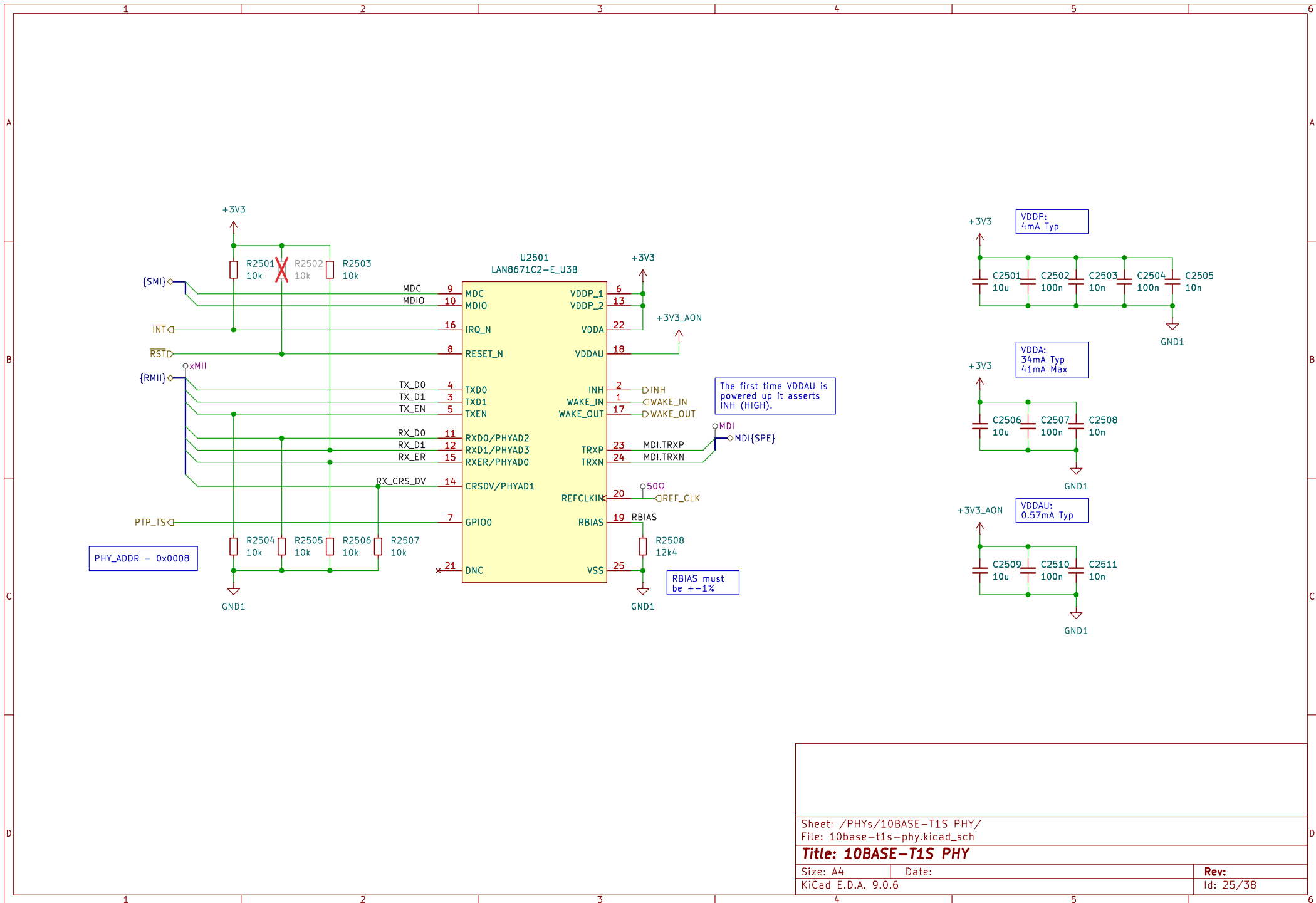
Sheet: /PHYs/1000BASE-T1 PHY 5/
File: 1000base-t1-phy.kicad_sch

Title: 1000BASE-T1 PHY

Size: A4
KiCad E.D.A. 9.0.6

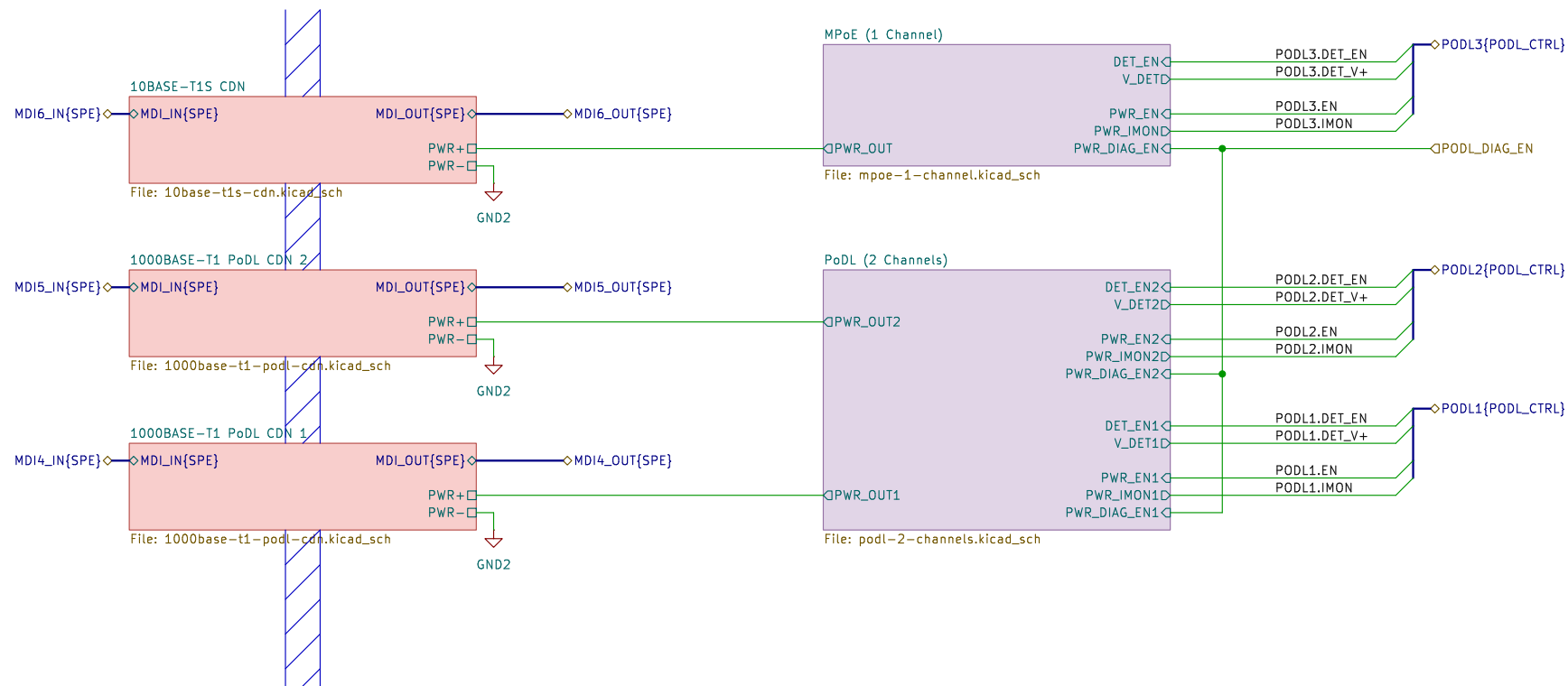
Date:

Rev:
Id: 24/38



Coupling-Decoupling Networks (CDNs)

Power Injection



Coupling-decoupling networks (CDNs) and power over datalines (PoDL).

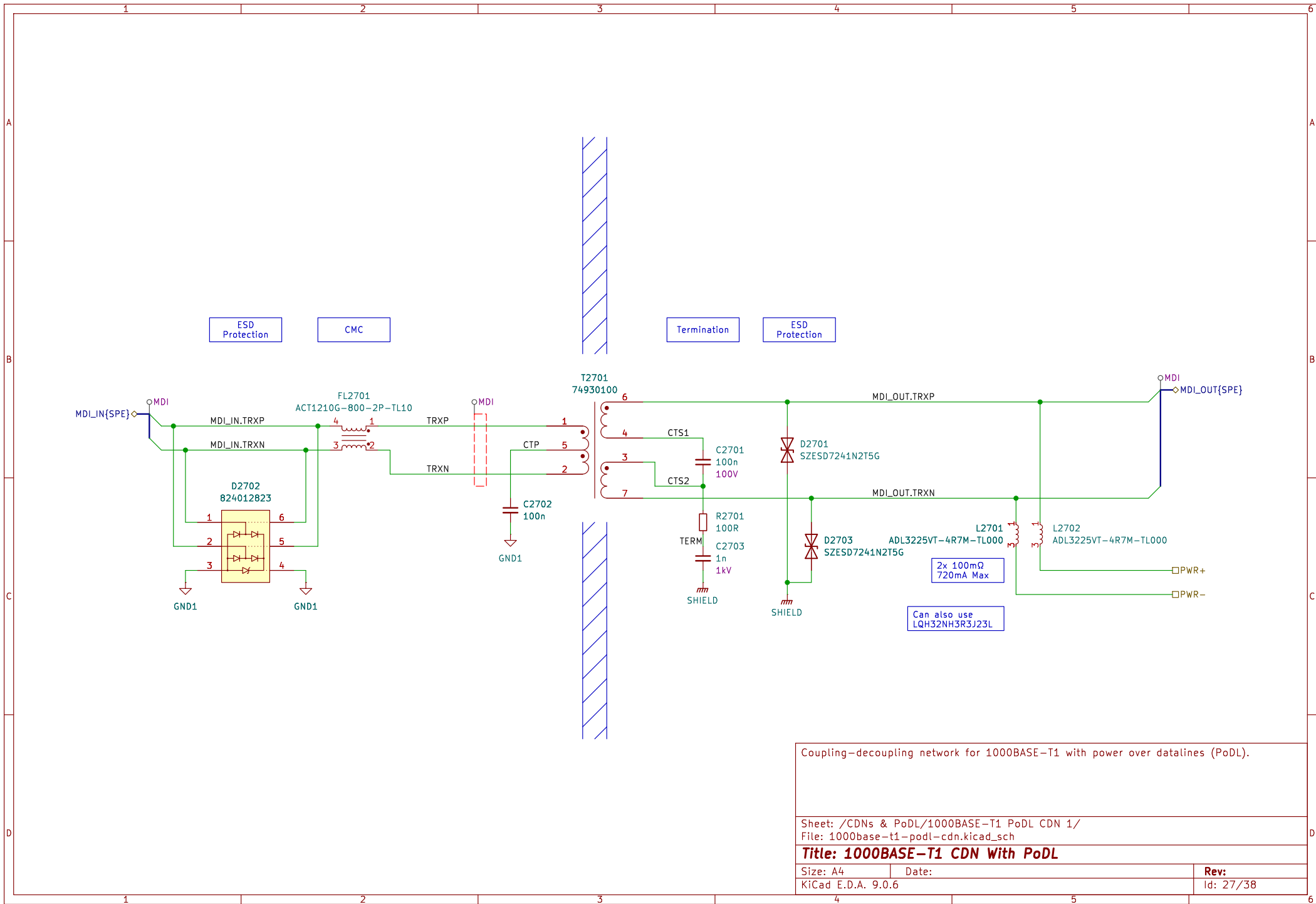
Sheet: /CDNs & PoDL/
File: cdn-podl.kicad_sch

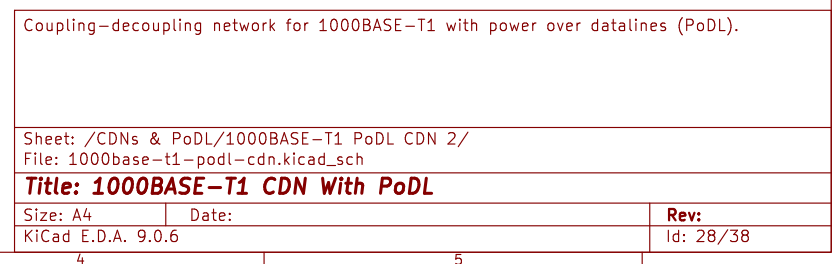
Title: CDNs and PoDL

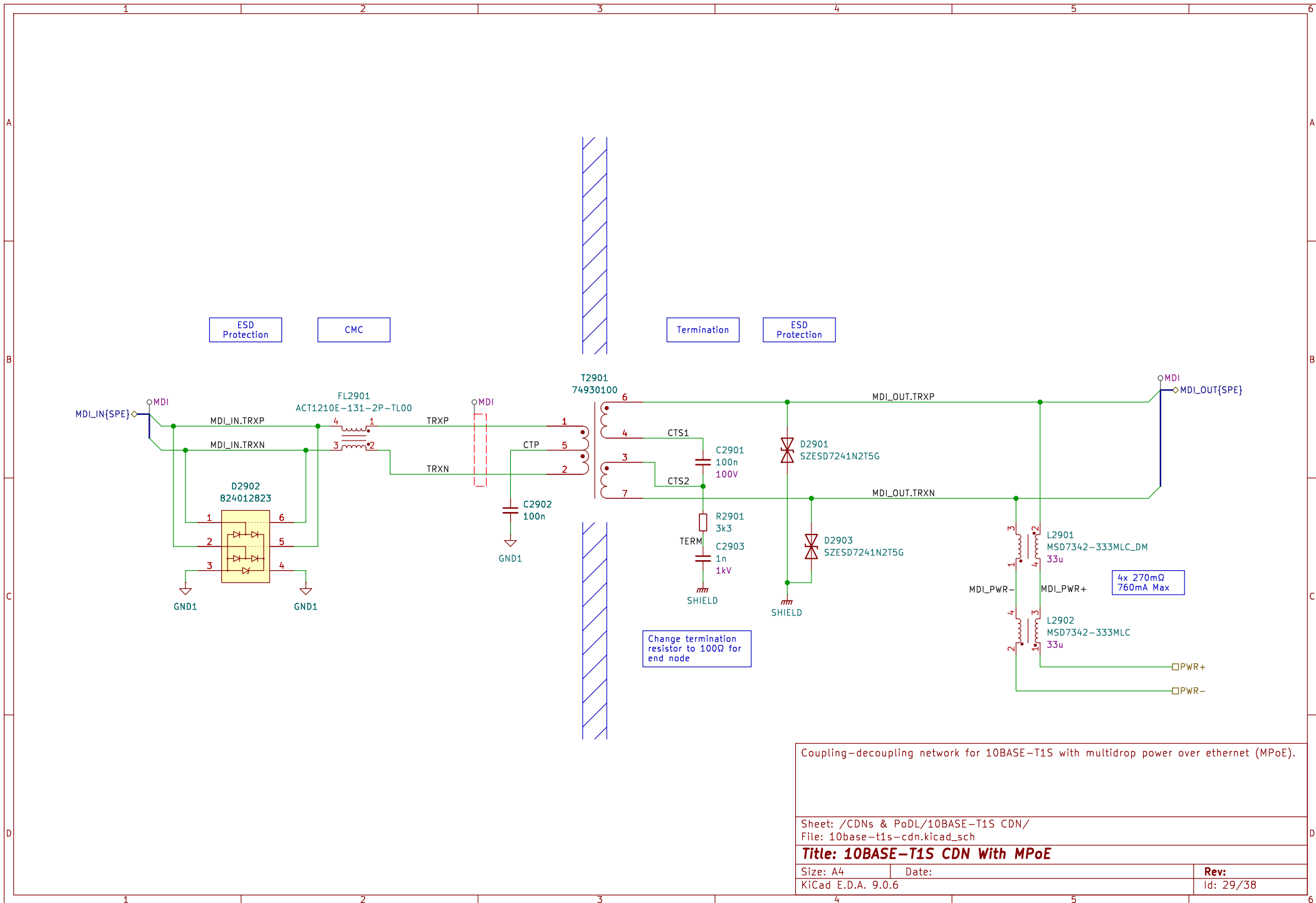
Size: A4
KiCad E.D.A. 9.0.6

Date:

Rev:
Id: 26/38







Coupling-decoupling network for 10BASE-T1S with multidrop power over ethernet (MPoE).

Sheet: /CDNs & PoDL/10BASE-T1S CDN/
File: 10base-t1s-cdn.kicad_sch

Title: 10BASE-T1S CDN With MPoE

Size: A4

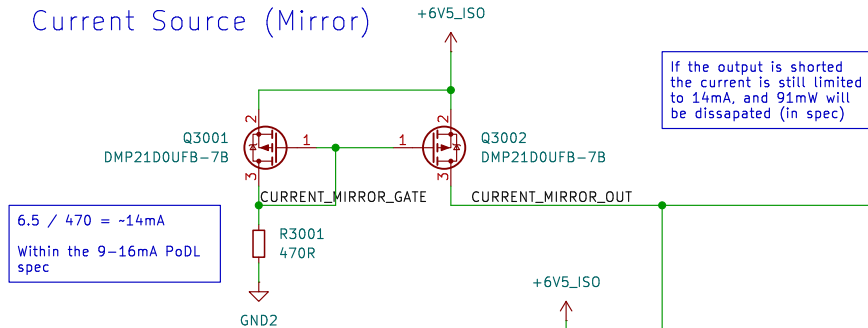
Date:

KiCad E.D.A. 9.0.6

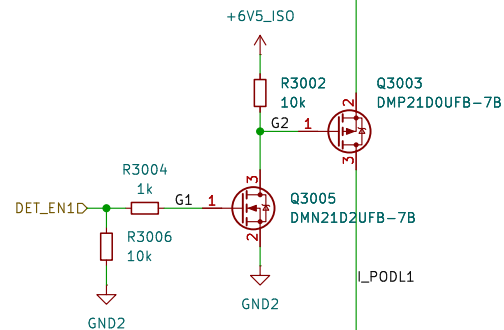
Rev:

Id: 29/38

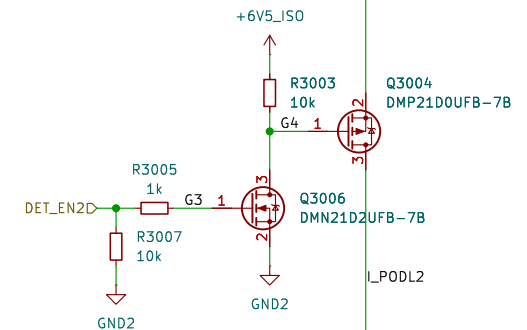
Current Source (Mirror)



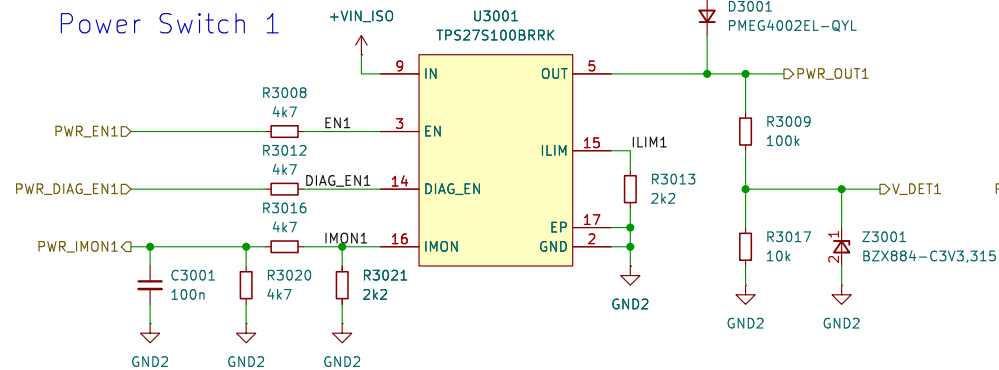
Current Switches



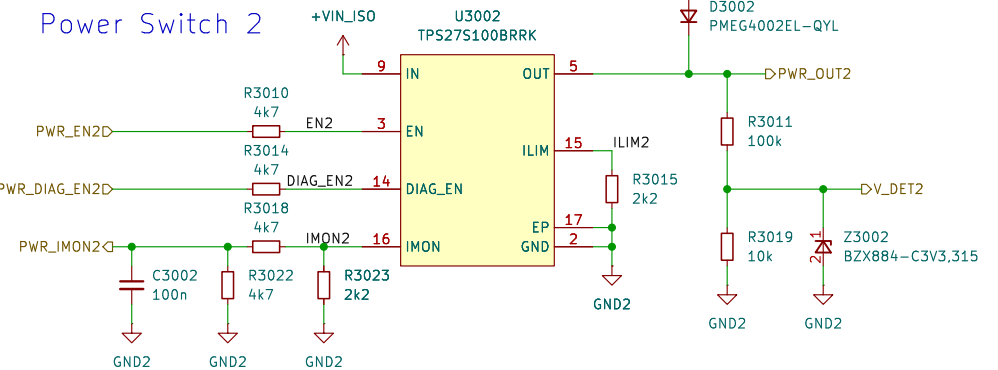
Each channel can be enabled individually to check for PoDL devices. Since the outputs share a current source, only one channel can be checked at a time.



Power Switch 1



Power Switch 2



Sheet: /CDNs & PoDL/PoDL (2 Channels)/
File: podl-2-channels.kicad_sch

Title: PoDL Detection, Switching and Monitoring

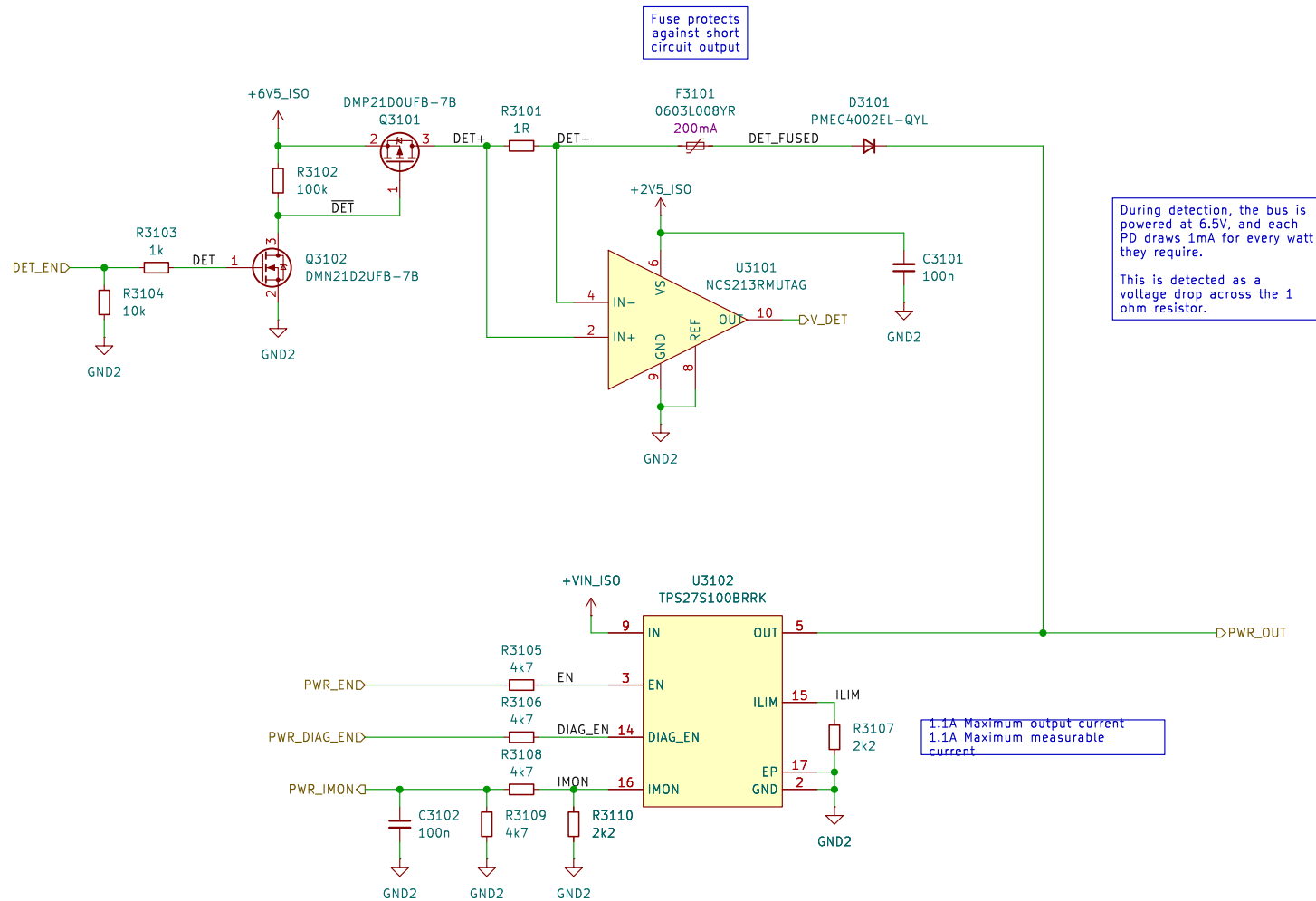
Size: A4

Date:

Rev:

KiCad E.D.A. 9.0.6

Id: 30/38



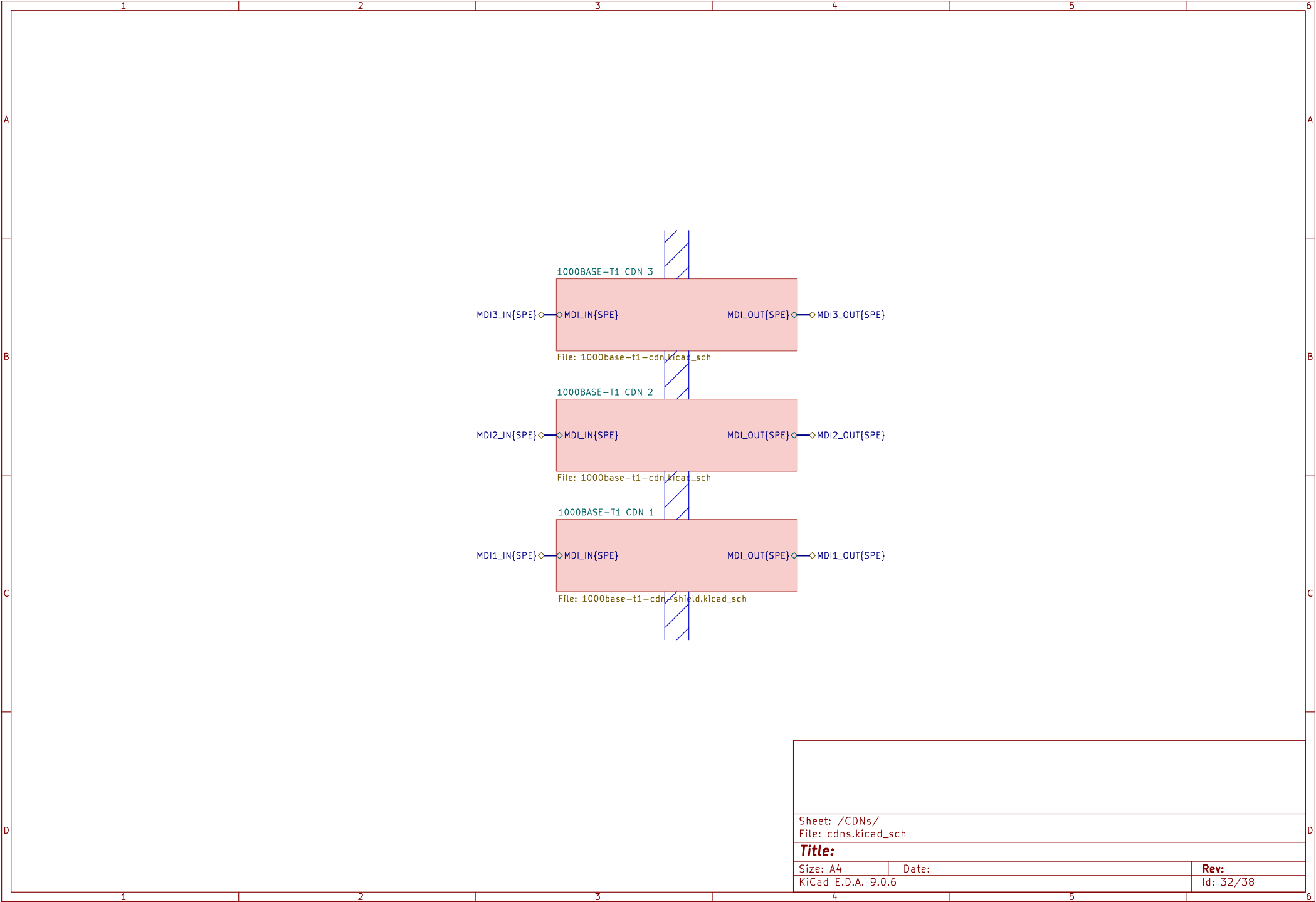
Fuse protects against short circuit output

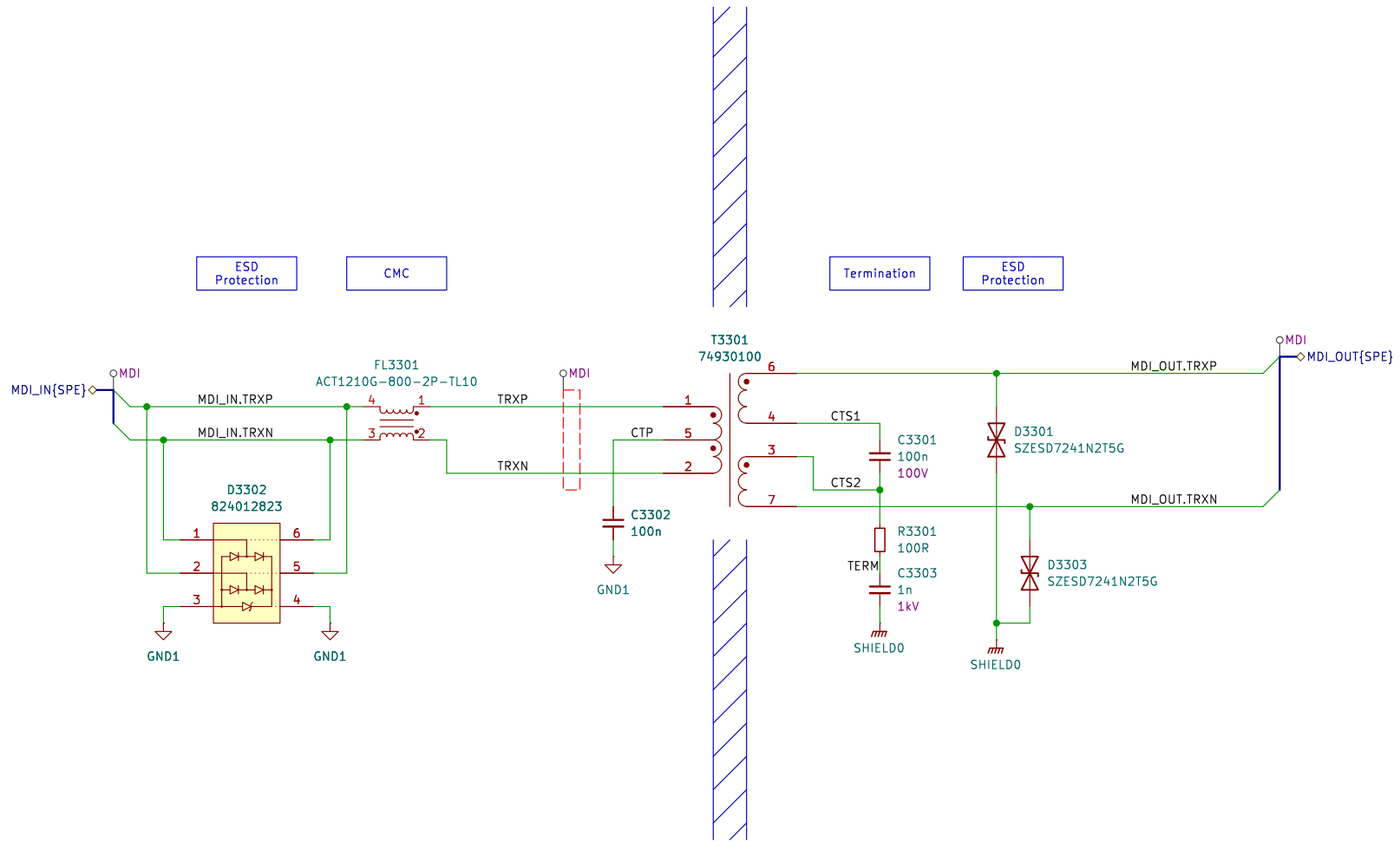
During detection, the bus is powered at 6.5V, and each PD draws 1mA for every watt they require.
This is detected as a voltage drop across the 1 ohm resistor.

1.1A Maximum output current
1.1A Maximum measurable current

IMON can go up to 4.75V, potential divider limits this to be within 2.5V ADC range.

$I_{OUT} = 0.28046 \cdot V_{MON}$





Coupling-decoupling network for 1000BASE-T1

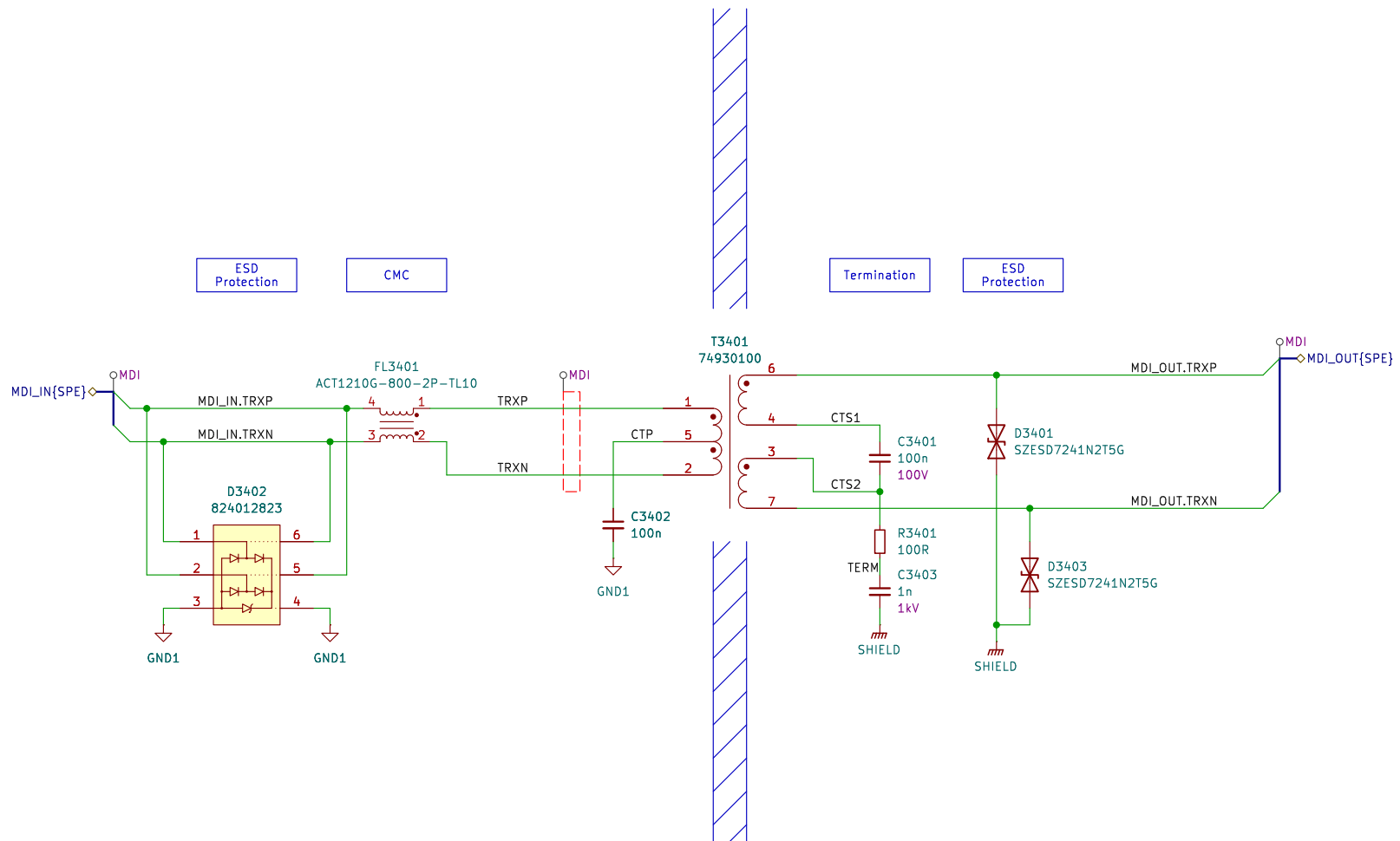
Sheet: /CDNs/1000BASE-T1 CDN 1/
File: 1000base-t1-cdn-shield.kicad_sch

Title: 1000BASE-T1 CDN

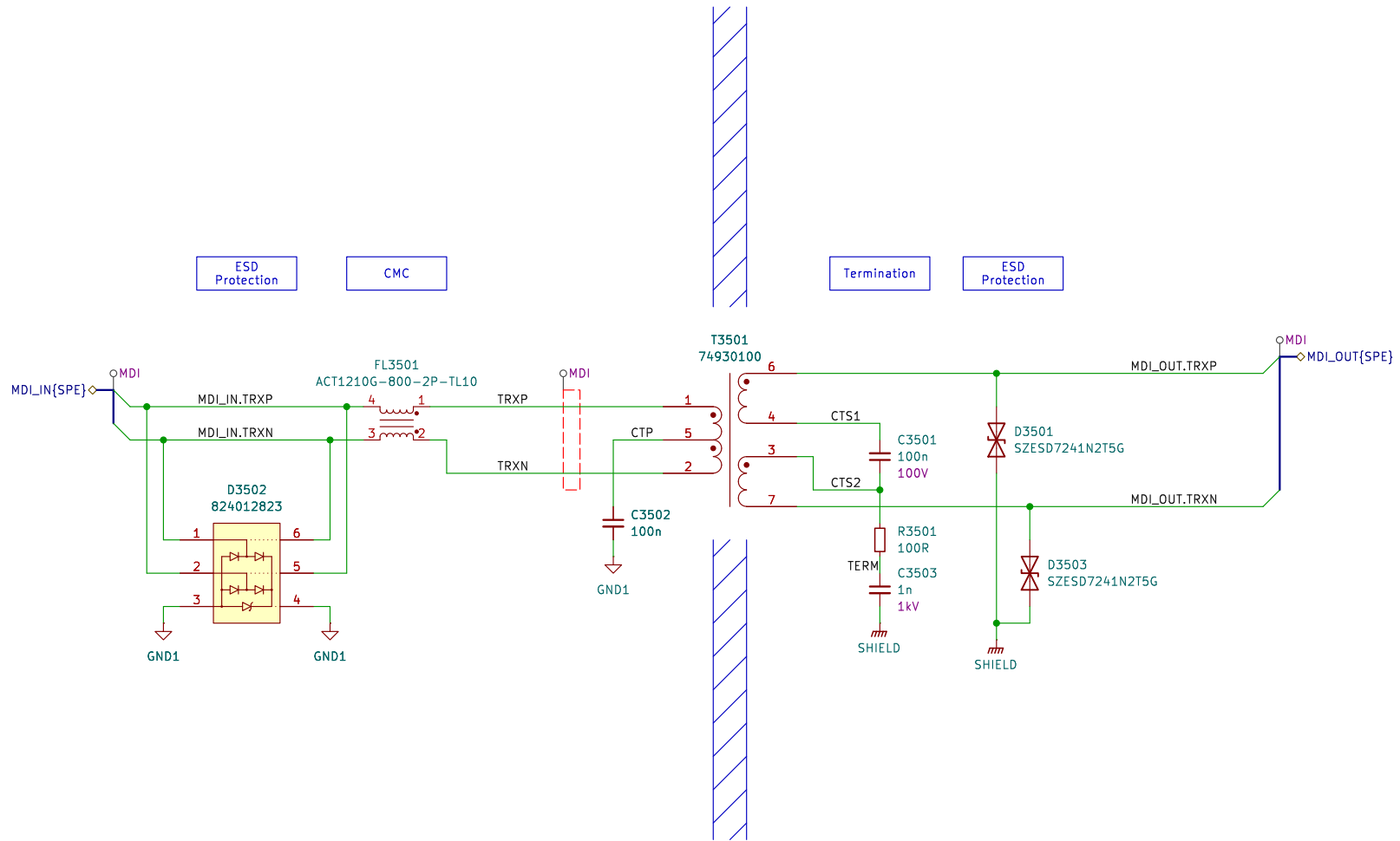
Size: A4
KiCad E.D.A. 9.0.6

Date:

Rev:
Id: 33/38



Coupling-decoupling network for 1000BASE-T1		
Sheet: /CDNs/1000BASE-T1 CDN 2/		
File: 1000base-t1-cdn.kicad_sch		
Title: 1000BASE-T1 CDN		
Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.6		Id: 34/38



Coupling-decoupling network for 1000BASE-T1

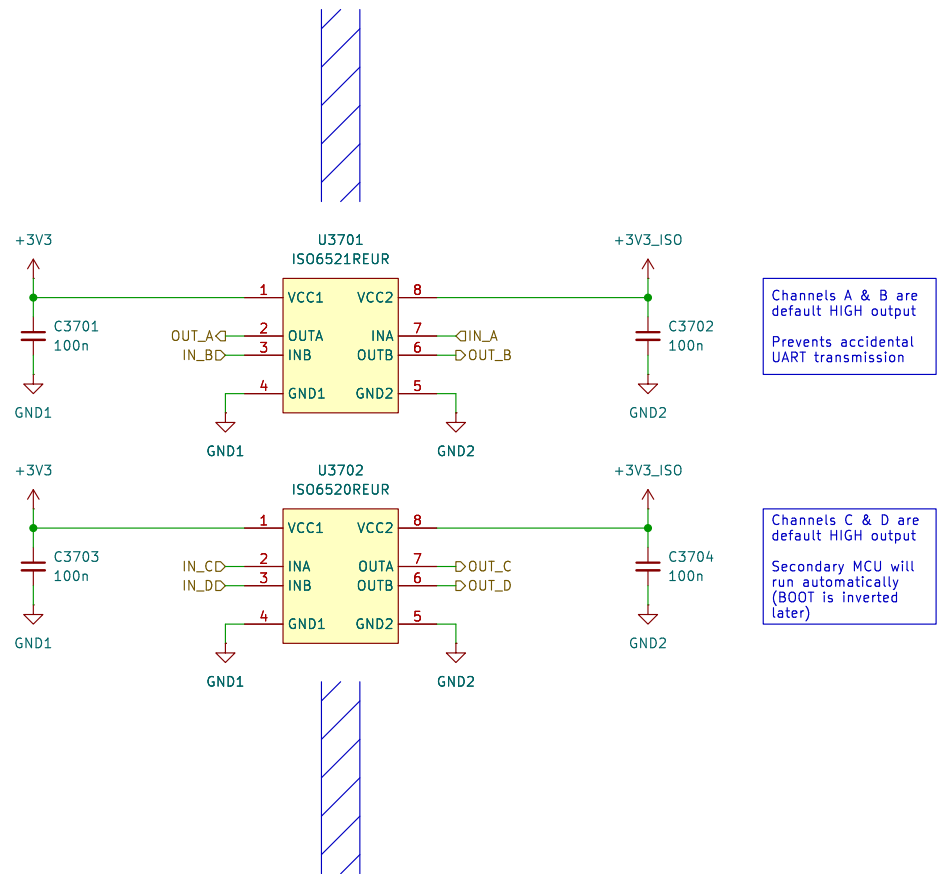
Sheet: /CDNs/1000BASE-T1 CDN 3/
File: 1000base-t1-cdn.kicad_sch

Title: 1000BASE-T1 CDN

Size: A4
KiCad E.D.A. 9.0.6

Date:

Rev:
Id: 35/38



Channels A & B are
default HIGH output
Prevents accidental
UART transmission

Channels C & D are
default HIGH output
Secondary MCU will
run automatically
(BOOT is inverted
later)

Sheet: /Isolation Crossing/		
File: isolation-crossing.kicad_sch		
Title: Isolation Crossing		
Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.6	Id: 37/38	

